Set 2: FASTQ quality improvement report

What this PDF contains

This report includes:

1. A boxplot of per-base Phred quality **before** processing.
2. A boxplot of per-base Phred quality **after** processing.
3. A short explanation of the decisions and how the quality improvement was implemented.

The required quality improvement consists of:

- removing all reads that contain at least one unknown base (`N`)
- trimming the same number of bases from the end of every read

Read data and compute quality before processing

Quality before processing

The boxplot below shows per-base Phred quality across reads, displayed every 5 positions.

Decision based on the BEFORE plot:
The BEFORE chart shows that the quality is very good and stable up to around 190-200 positions. The median is usually around 37-39. After that, there is a clear decline and a larger spread of quality values. The final positions, especially around 230-251, contain many bases with quality around Q20 and below, and the last positions drop even closer to Q15. Therefore, trimming was done by removing about 50 bases from the end, leaving the first 200 bases.

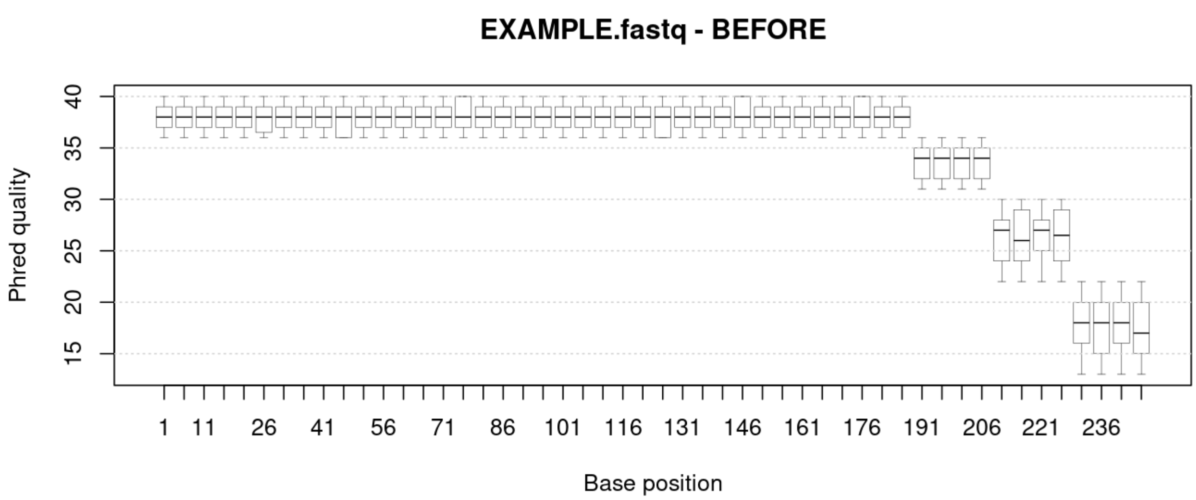

Figure 1: EXAMPLE.fastq - BEFORE: per-base Phred quality boxplot

Implementation of quality improvement

Step 1: filtering

All reads containing at least one unknown base (`N`) are removed.

Step 2: trimming

A fixed number of bases is trimmed from the end of every read.
Here, I trim **k = 50** bases from the end of each read. The same value is used for all reads and was selected based on the BEFORE quality plot.

Apply filtering and trimming

```
# 1) Remove reads containing N
n_count <- letterFrequency(seq_raw, letters = "N")[, 1]
keep_noN <- (n_count == 0)
fastq_noN <- fastq_raw[keep_noN]

n_removed_N <- sum(!keep_noN)

# 2) Trim a fixed number of bases from the end
end_pos <- width(sread(fastq_noN)) - k_trim
keep_len <- end_pos >= 1

fastq_trim <- narrow(
  fastq_noN[keep_len],
  start = 1,
  end = end_pos[keep_len]
)

n_after_trim <- length(fastq_trim)
qual_after <- as(quality(fastq_trim), "matrix")
```

Quality after processing

Short summary

- Input reads: **120**
- Reads removed due to `N`: **12**
- Fixed trimming length from the end: **50** bases
- Reads after processing: **108**

The plots saved for the portfolio website are:

- `figures/fastq_before.png`
- `figures/fastq_after.png`

Executable R script

The extracted `.R` script accepts a FASTQ file name as a command-line argument or asks for it interactively if it is missing. It removes reads containing at least one `N`, trims the same number of bases from the end of every read, and saves two output files: the processed FASTQ file and the final quality boxplot as a PNG image. The user can choose the trim length, plot step and output compression.

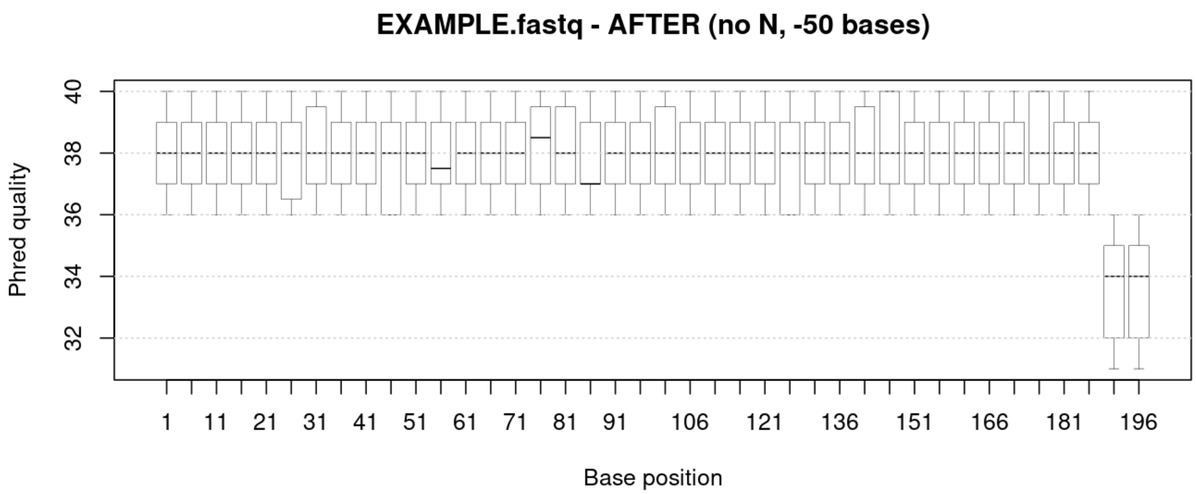

Figure 2: EXAMPLE.fastq - AFTER: after removing reads with N and trimming a fixed number of bases from the end